Numerical Analysis I

Quiz 10A Name:_____

Consider the nonlinear model

$$y(t;c) = \frac{c}{1+ct}$$

and the two data points

$$(t_1, y_1) = (0, 2), \qquad (t_2, y_2) = (1, 1).$$

We wish to find the parameter c that minimizes

$$f(c) = \frac{1}{2}\sum_{i=1}^{2} \left(y(t_i;c) - y_i\right)^2 .$$

Starting from $c^{(0)} = 1$, perform one iteration of the Gauss–Newton method. Show your work and give the updated parameter $c^{(1)}$.

The residual vector is

$$r(c) = \begin{pmatrix} c-2 \\ \dfrac{c}{1+c} - 1 \end{pmatrix}.$$

The Jacobian is

$$J(c) = \begin{pmatrix} 1 \\ \dfrac{1}{(1+c)^2} \end{pmatrix}.$$

At $c^{(0)} = 1$, we have

$$r(1) = \begin{pmatrix} -1 \\ -\dfrac{1}{2} \end{pmatrix}, \qquad J(1) = \begin{pmatrix} 1 \\ \dfrac{1}{4} \end{pmatrix}.$$

The Gauss–Newton step p satisfies

$$J(1)^T J(1) p = -J(1)^T r(1).$$

Substituting,

$$\left(1 + \frac{1}{16}\right) p = 1 + \frac{1}{8},$$

so

$$\frac{17}{16} p = \frac{9}{8}, \qquad p = \frac{18}{17}.$$

Therefore, the updated parameter is

$$\boxed{c^{(1)} = c^{(0)} + p = 1 + \frac{18}{17} = \frac{35}{17}.}$$

Trying this in python and doing more steps we get

k	c	$\|\|r\|\|$	$\|p\|$
0	1.0000000000	1.1180339887e+00	1.0e+00
1	2.0588235294	3.3217300588e-01	2.3e-02
2	2.0352108419	3.3134259375e-01	5.4e-04
3	2.0357565261	3.3134214970e-01	1.2e-05
4	2.0357438234	3.3134214946e-01	2.9e-07
5	2.0357441190	3.3134214946e-01	6.8e-09
6	2.0357441121	3.3134214946e-01	1.6e-10
7	2.0357441123	3.3134214946e-01	3.7e-12
8	2.0357441123	3.3134214946e-01	8.6e-14

